Code No: 134AK **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech II Year II Semester Examinations, July/August - 2021
COMPUTER ORGANIZATION
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1. Draw the block diagram of the microprogram sequencer for a control memory and explain its operations in detail. [15]

2. A digital computer has a memory unit with a capacity of 16384 words 40 bits per word. The instruction code format consists of six bits for the operation part and 14 bits for the address part. Two instructions are packed in one memory word, and a 40-bit instruction register IR is available in the control unit. Formulate a procedure for fetching and executing instructions for this computer. [15]

3. Explain the 8086 Processor Architecture. [15]

4. Draw and discuss a typical minimum mode of 8086 system. [15]

5. Elaborate the stack structure of the 8086. [15]

6. Describe the procedure for coding the intersegment and intersegment jump and call instructions. [15]

7. How many characters per second can be transmitted over a 1200 baud line in each of the following modes?
   a) Synchronous serial transmission
   b) Asynchronous serial transmission with two stop bits.
   c) Asynchronous serial transmission with one-stop bits [5+5+5]

8.a) Multiprocessing can improve performance by decomposing a program into parallel executable tasks. Explain how this is achieved.
   b) What is associative memory? Why is it faster than the main memory? Explain its hardware organization in detail. [8+7]

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